

Department of Mathematics
University of Michigan
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Research Interests: I am interested in mathematical questions arising in quantum information theory, including problems in quantum algorithms, quantum error correction, and mathematical physics.

Employment:

2026– **Postdoctoral Assistant Professor**
University of Michigan, Ann Arbor, MI.

Short-Term Positions:

2023 **Summer Intern**
Los Alamos National Laboratory, Los Alamos, NM.

Education:

2020–2026 **Ph.D. in Mathematics**
University of Maryland, College Park.
2018–2020 **M.S. in Mathematics**
Kansas State University.
2013–2017 **B.S. in Economics & Mathematics**
Lahore University of Management Sciences (LUMS).

Preprints:

1. **Junaid Aftab.** *A Cohomological Characterization of the Clifford Hierarchy.* [arXiv:2610.06699](https://arxiv.org/abs/2610.06699).
2. **Junaid Aftab**, Yuehaw Khoo, and Haizhao Yang. *Non-Uniform Quantum Fourier Transform.* [arXiv:2602.13472](https://arxiv.org/abs/2602.13472).
3. **Junaid Aftab**, Christoph Schwab, Haizhao Yang, and Jakob Zech. *Quantum Circuit Encodings of Polynomial Chaos Expansions.* [arXiv:2506.01811](https://arxiv.org/abs/2506.01811).
4. **Junaid Aftab**, Dong An, and Konstantina Trivisa. *Multi-product Hamiltonian simulation with explicit commutator scaling.* [arXiv:2403.08922](https://arxiv.org/abs/2403.08922).

Publications:

5. **Junaid Aftab** and Haizhao Yang. *Approximating Korobov Functions via Quantum Circuits.* *Communications in Mathematical Sciences* 23(8), pp. 2077–2101 (2025).
6. **Junaid Aftab** and Adam Zaman Chaudhry. *Analyzing the quantum Zeno and anti-Zeno effects using optimal projective measurements.* *Scientific Reports* 7, 11766 (2017).

Awards & Fellowships:

2024–2026 MathQuantum NSF RTG Fellowship, University of Maryland, College Park.
2024 Herbert A. Hauptman Summer Fellowship, University of Maryland, College Park.
2020–2022 Dean’s Fellowship, University of Maryland, College Park.
2017 NMF Gold Medal, Lahore University of Management Sciences.
2013–2017 Dean’s Honour List, Lahore University of Management Sciences.

Talks:

Sep. 2026 *Quantum Simulation via Multi-Product Formulas*
Purdue–Michigan Quantum Algorithms Seminar
May 2026 *Non-Uniform Quantum Fourier Transform*
MathQuantum RTG Symposium, University of Maryland
May 2025 *Multi-Product Hamiltonian Simulation with Explicit Commutator Scaling*
MathQuantum RTG Symposium, University of Maryland
Mar. 2025 *Hyperbolic Band Theory*
Research Interaction Team (RIT) on Geometry and Physics, University of Maryland

- Sep. 2024 *Multi-Product Hamiltonian Simulation with Explicit Commutator Scaling*
TQC Conference, Okinawa Institute of Science and Technology
- Mar. 2023 *Topological Quantum Computation*
Research Interaction Team (RIT) on Geometry and Physics, University of Maryland
- May 2022 *Homological Product Codes*
Quantum Error Correction Reading Group, University of Maryland

Teaching:

- 2026– **University of Michigan, Instructor**
- MATH 214: Applied Linear Algebra, Fall 2026
- 2022–2026 **University of Maryland, Instructor**
- MATH 141: Calculus II, Summer 2026
 - MATH 120: Elementary Calculus, Summer 2024
 - MATH 141: Calculus II, Summer 2022
- 2020–2023 **University of Maryland, Graduate Teaching Assistant**
- MATH 240: Introduction to Linear Algebra, Fall 2023
 - MATH 140: Calculus I, Fall 2022
 - MATH 135: Discrete Mathematics for Life Sciences, Fall 2021
 - MATH 141: Calculus II, Spring 2021, Spring 2023
 - MATH 120: Elementary Calculus, Fall 2020
- 2018–2020 **Kansas State University, Graduate Teaching Assistant**
- MATH 340: Elementary Differential Equations, Fall 2019, Spring 2020
 - MATH 220: Analytic Geometry and Calculus I, Fall 2018, Spring 2019

Conferences:

- Jun. 2025 Topological Phases via Homotopy and Operator Algebras, Harvard University.
- Aug. 2024 C*-Algebraic Quantum Mechanics and Topological Phases of Matter, CU Boulder.
- Jul. 2024 Groundwork for Operator Algebras Lecture Series, Institute for Pure & Applied Math.
- Jul. 2023 PCMI Graduate Summer School, Park City Mathematics Institute (PCMI).

Mentoring:

- 2025 **University of Maryland, Research Mentor**
- Ava Petusky, Spring 2025: Quantum Simulation
- 2022–2026 **University of Maryland, Directed Reading Program**
- Mina Ricotti, Spring 2026: Manifolds, Fundamental Group and Homology
 - Kelin Zhu, Fall 2025: K -Theory
 - Nashita Bhuiyan, Spring 2024: Learning Theory: PAC-Learning & VC Dimension
 - Koran Bailey, Spring 2023: Grover's Search Algorithm & Random Walks
 - Matthew Cimerola, Fall 2022: Neural Networks & Their Applications

Service:

- 2024–2025 **MathQuantum RTG Outreach, University of Maryland**
- Maryland Day 2025, Halloween Science Festival 2024
- 2024 **Course Staff, University of Maryland**
- AMSC 698: Mathematics of Quantum Information
- 2024 **Guest Lecturer, University of Maryland**
- Delivered two lectures for a graduate course on differential geometry

Software:

- Languages Python, Julia, MATLAB, Mathematica, Lean, GAP, $\text{\LaTeX}$.
- Libraries Qiskit, PyTorch, NumPy, SciPy, SymPy, Pandas, Scikit-Learn, SageMath, Matplotlib.
- Certification IBM Certified Qiskit Associate Developer, 2022. [Badge](#).